Inductive Interface

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Microsoft- http://msdn.microsoft.com/library

Books and References

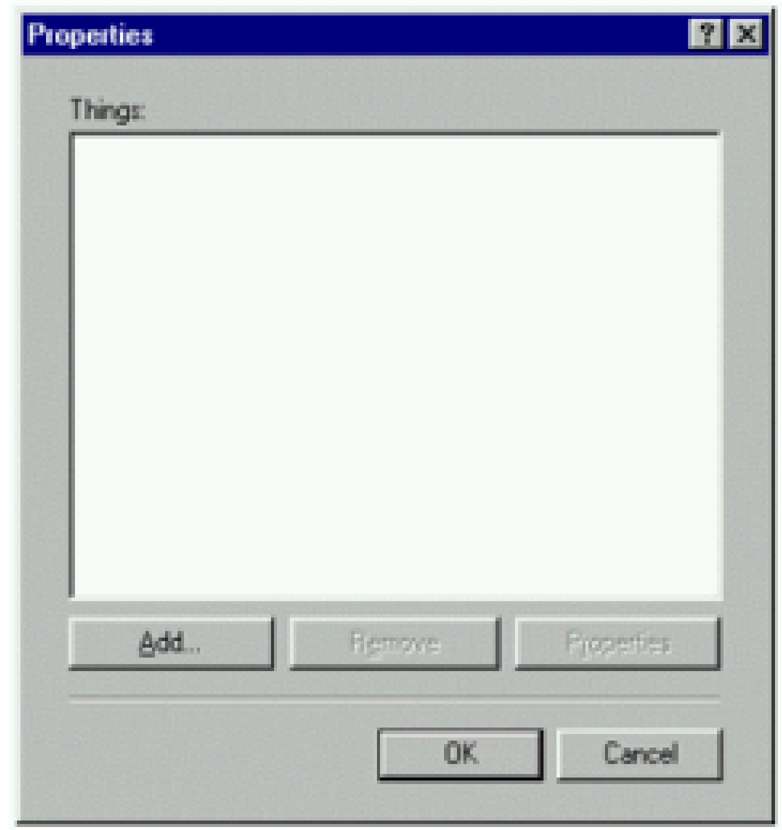
- Shneiderman and Plaisant, "*Designing the User Interface*",
- Stone *et. al.*, "*User Interface Design and Evaluation*",
- Benyon *et. al.*, "*Designing Interactive Systems: People, Activities, Contexts, Technologies*"
- References will be given for each lecture

What is Inductive User Interface?

- The Inductive User Interface model starts from the premise that software is hard to use.
- The inductive interface model suggests that software can be made easier and simpler by:
  - Breaking down features into screens that are easy to explain
  - Focusing each screen on a single task
  - Suiting a screen's contents to the task
  - Making it obvious how to complete a task using the controls on the screen

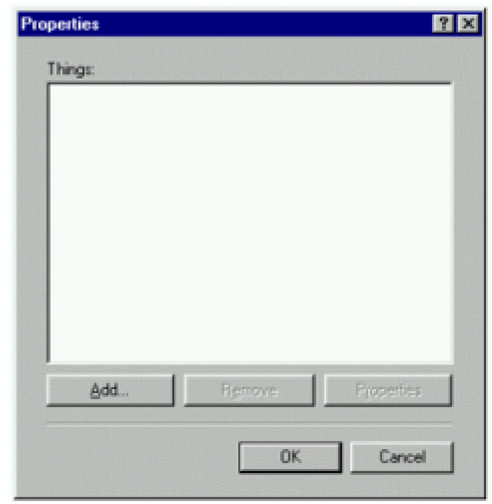
Deductive vs. Inductive

- Most elements in software today require the user to study them and deduce their behavior.
- For experienced computer users – it is obvious.
- However, none of this behavior is explicitly stated in the dialog itself!



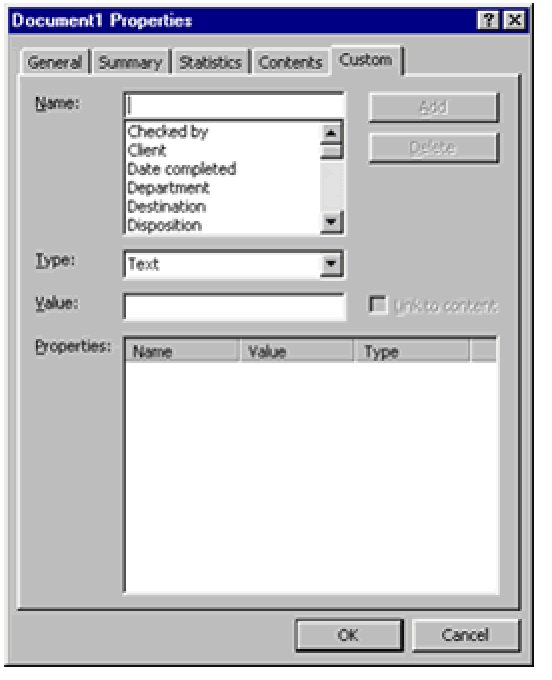
Deductive vs. Inductive – cont.

- But from the view point of casual user:
  - "What am I supposed to do with this?"
  - When the dialog appears, the user must stop and figure out what to do next.



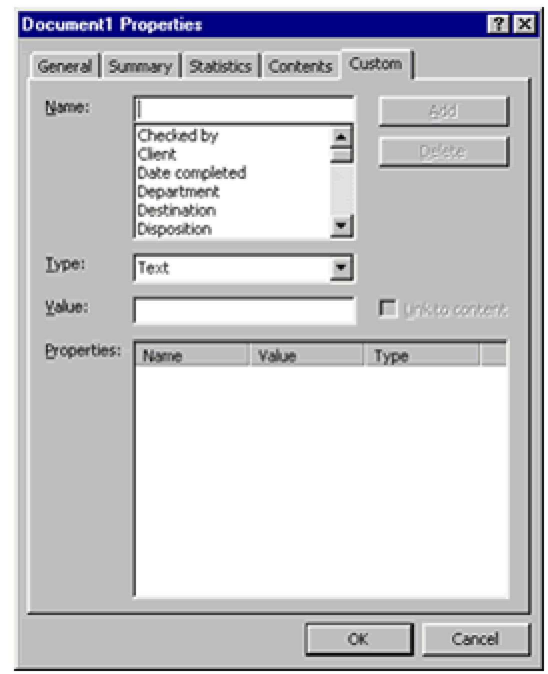
Deductive vs. Inductive – cont.

- You probably have dozens of examples of deductive user interfaces sitting on your desktop right now.
- Here's one selected from Microsoft Word.



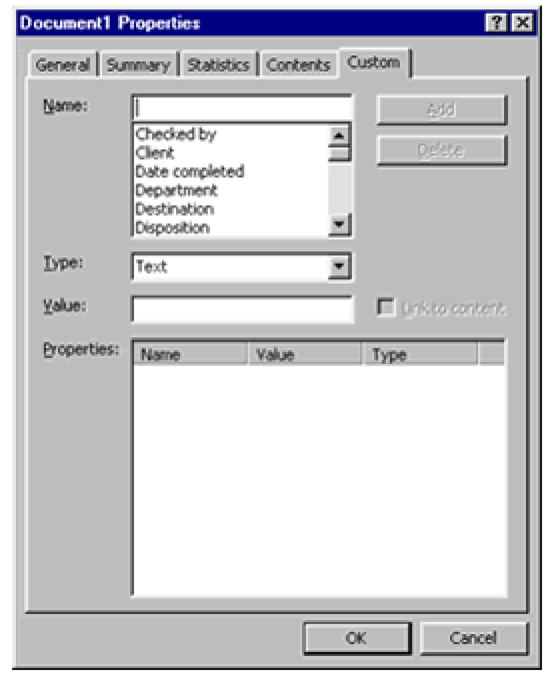
Deductive vs. Inductive – cont.

- In order to fully understand this dialog box, one must:
  - Deduce the purpose and correct use of the Name, Type, Value and Properties boxes.
  - Deduce the significance of the disabled **Add** and **Delete** buttons.
  - Deduce the impact, if any, of clicking another tab from within the **Custom** tab.



Deductive vs. Inductive – cont.

- Deduce the consequences of pressing **OK** or **Cancel**.
- After we have successfully deduced all of those items, perhaps we'll be able to deduce the purpose of the screen, too.



The Consequence: Software is hard to use

- This conclusion is drawn from usability testing, subjective evidence, and personal experiences of software designers.

- The concept of inductive interface was created by conducting research, making educated guesses as to what makes current software hard to use, and then proposing solutions.

Software products are hard to use for the following general reasons:

- Users don't seem to construct an adequate mental model of the product (conceptual model).
- Even many long-time users never master common procedures (Studies show that many users are confused even by basic operations in software).
- Users must work hard to figure out each feature or screen (In a way, software designers create programs for themselves).
  - Most programs give a set of controls, but leave it to the user to deduce the page's purpose and how to use it.

Outcome of these problems

- Users **are distracted** from their goals whenever they must figure out the purpose of a screen and how to use it.

- This ultimately represents a **cost in time** and **user satisfaction**.

- What's worse, users pay this **cost over and over again** as they puzzle over the interface each time they use a feature.

The idea

- A well-designed inductive interface helps users answer two fundamental questions they face when looking at a screen:

- What am I supposed to do now?
- Where do I go next?

- Users must be able to find a feature every time they need it, and must be able to use that feature every time they want to use it.

The fundamental premise

- Software that uses inductive interface answers these questions by starting with a fundamental premise:

A screen with a single, clearly stated, explicit purpose is easier to understand than a page without such a purpose.

- If the screen is easier to understand, it will be easier for the user to know what to do and where to go next.

Process – a single task

- Many tasks require users to navigate through a series of screens.
- To provide this ability, inductive interface defines a navigational concept called a **process**, a screen or series of screens that perform a task.
- A process acts as a sort of navigational subroutine.
- Then on the last page they can click a "Done" button to quickly return to the page where they began the process

Steps Towards Inductive Interface

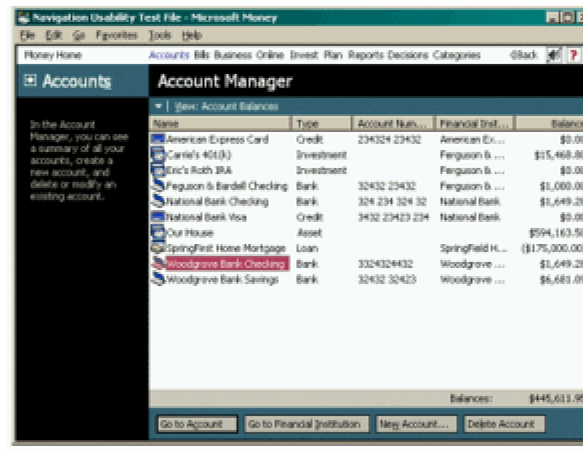
1. Focus each page on a single task.
2. State the task.
3. Make the page's contents suit the task.
4. Offer links to secondary tasks.

Step one:
Focus each page on a single task.

- This idea sounds simple, though hardly ever used.
- Each screen should be focused on a single task, called the screen's primary task.
- The primary task can be either specific or open ended.
- For example, in a personal finance program:
  - A specific task might be "Select the bill you want to pay,";
  - An open-ended task might be "Review the performance of your investments."
- The task should be something users might think to do, preferably described in their own words.

Focus each page on a single task – cont.

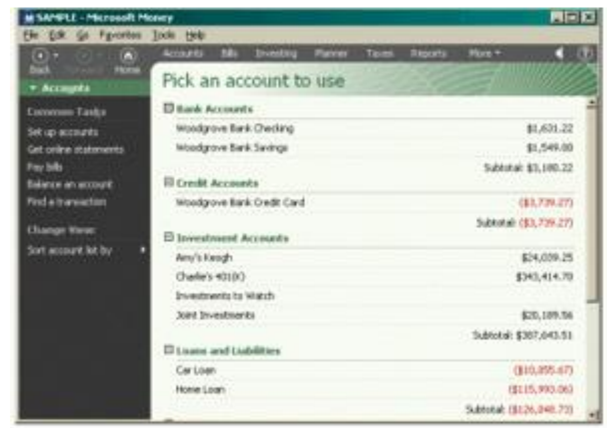
- In **Microsoft Money 99**, users often performed a large variety of tasks on a single screen.
- The example screen shown here groups a common task - navigating to an account, as well as infrequent tasks like creating and deleting accounts.
- None of these specific tasks is directly expressed in the screen's title, "**Account Manager**".
- The user must deduce the purpose of the screen and how to use it.



Multiple tasks on a single screen

Focus each page on a single task – cont.

- Money 2000, which uses inductive interface, offers a nearly identical set of account-related features, but provides them on two separate screens.
- Money 2000 is a web style application.
- This example shows the first of these screens, which is focused entirely on getting the user to pick an account.



Focus each page on a single task – cont.

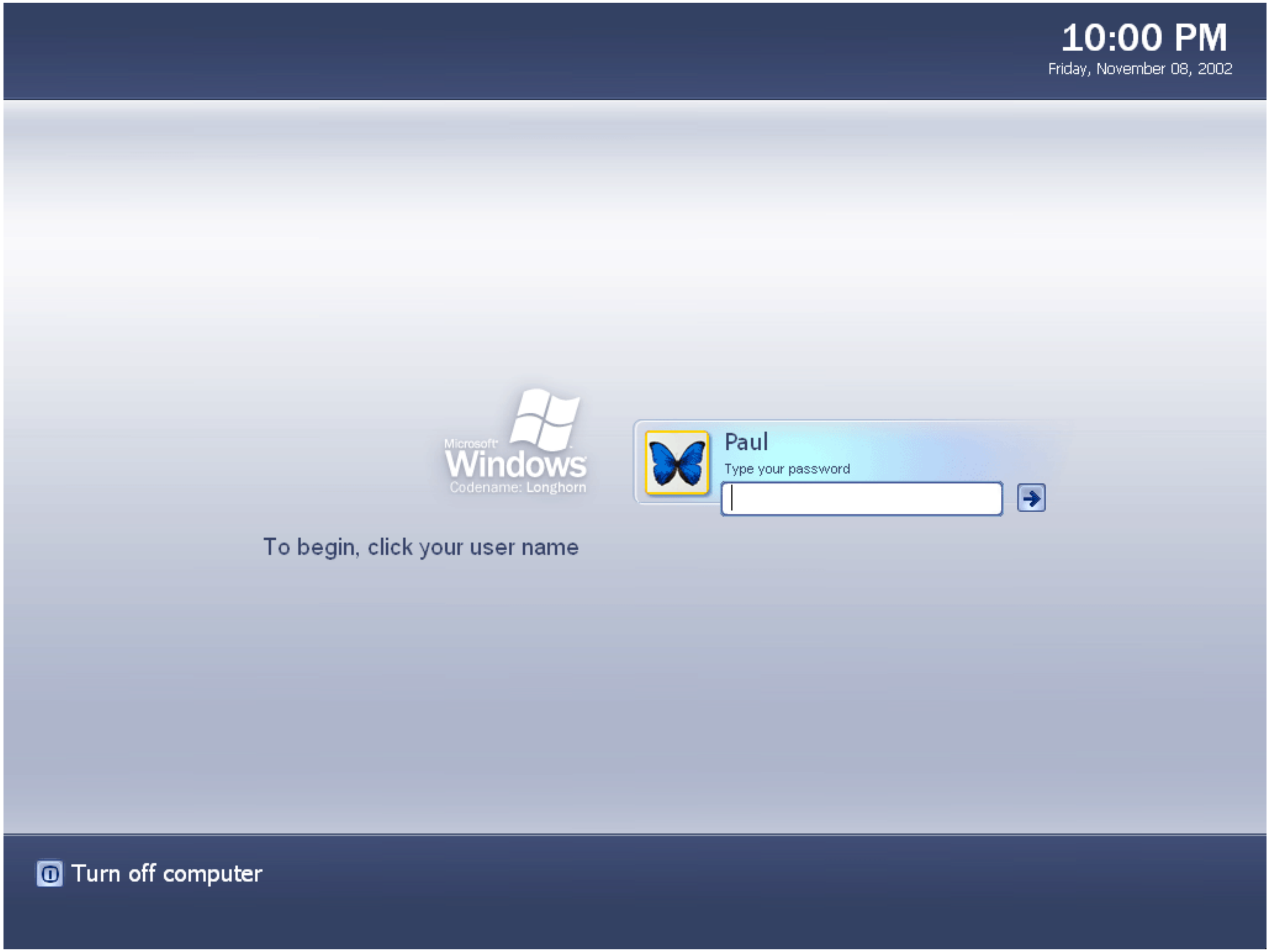
- To add or remove an account in Money 2000, users navigate to a second screen focused on setting up accounts.
- The purpose of each screen is clearer in Money 2000's inductive interface.
- Each screen has more space to devote to fulfilling its purpose.

What is a single task?

- How do you know if a screen is really focused on just one task?

- Here is a rule of thumb:

A screen is focused on one purpose if the designer can express that purpose with a concise, meaningful, and natural sounding screen title.

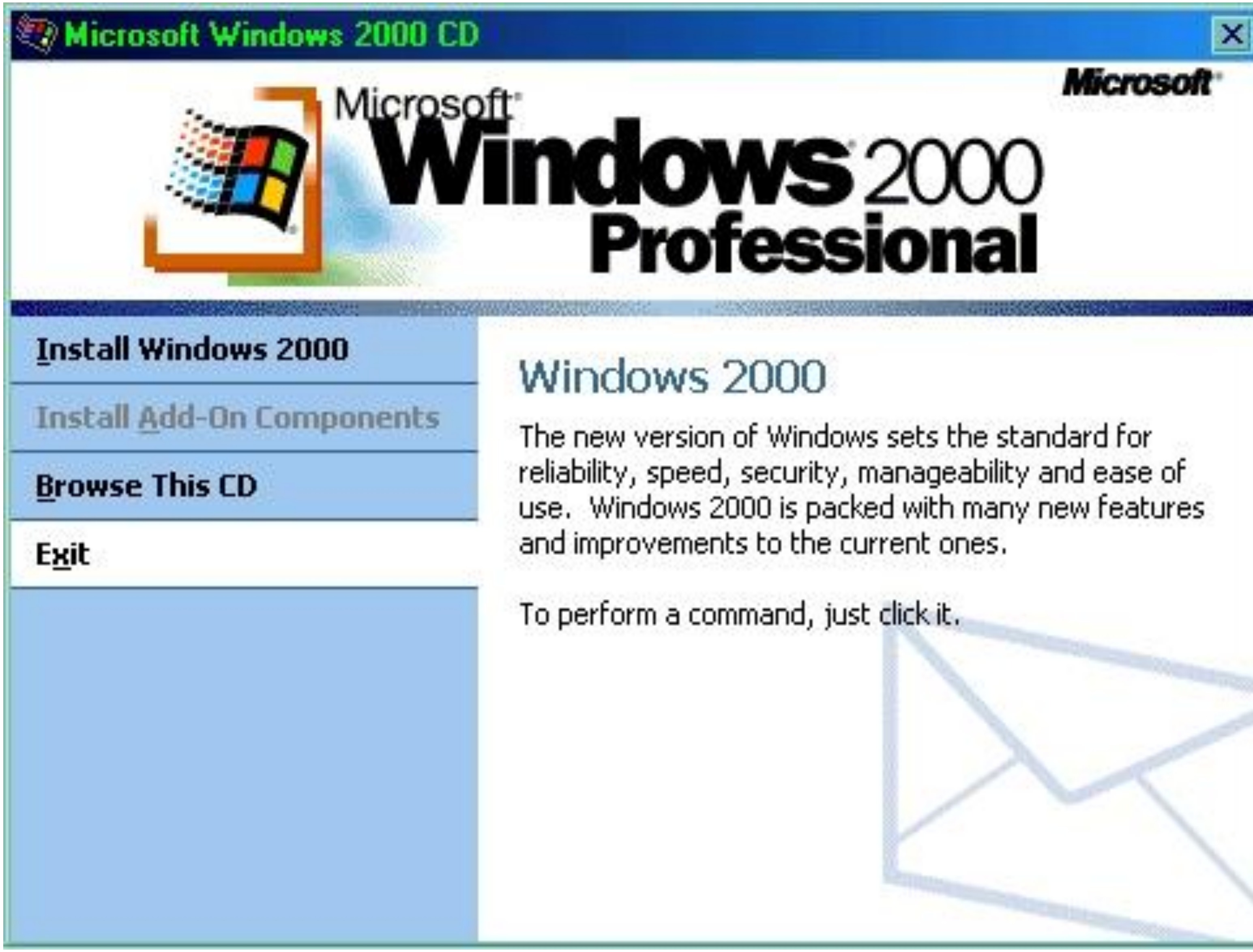


Step one: focus each page on a single task.

Step two:
State the task

- Each screen should be titled with a concise and explicit statement of its primary task.

- This can be a direct instruction ("Select the account you want to balance") or a question you want the user to answer ("Which account do you want to balance?").



Windows 2000 setup screen

Windows XP setup screen

Usable screens have clear titles

- The abstract title "Account Manager" in Money 99 was given to this page in an attempt to capture a variety of tasks.

- The main purpose of the "Account Manager" page was simply choosing an account.

- What is a bad title:
  - Names that are deliberately vague, such as "Settings",
  - Coined buzzwords, like "QuickStep".
  - Jargon that reveals implementation details ("Database Compaction").
  - Using nouns that don't express clearly the action the user wants to accomplish ("Accounts").

Usable screens have clear titles – cont.

- Screen titles and other names and words are *often not determined until* near the *end of the design process*.
  - At that point, there is **no recourse** if a good name can't be found, and so the team may have to settle for names that are not clear.

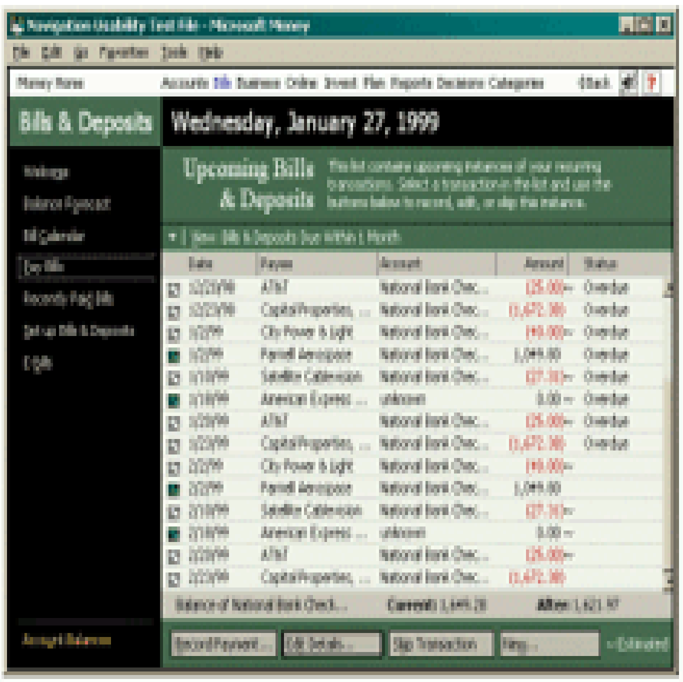
- In the inductive interface model, the designers choose the screen titles in the **earliest stages of the design process**.

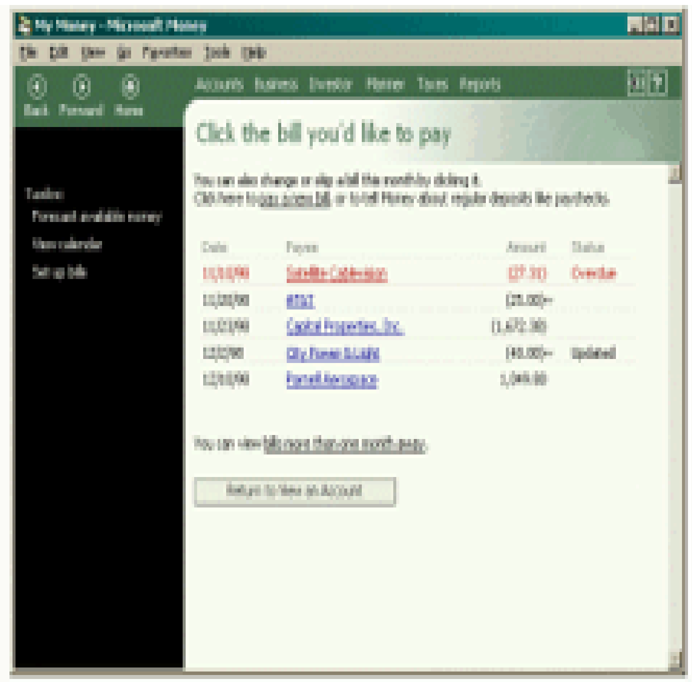
- Screen functions and titles should focus on the **most common tasks** performed by customers.
  - Designers are often tempted to provide enormous amounts of functionality – adds complexity.

- If no suitable title can be found, the feature is redesigned. If no design permits a clear and concise title - if there is no way to explain the feature - the designers might abandon the feature.

Example:

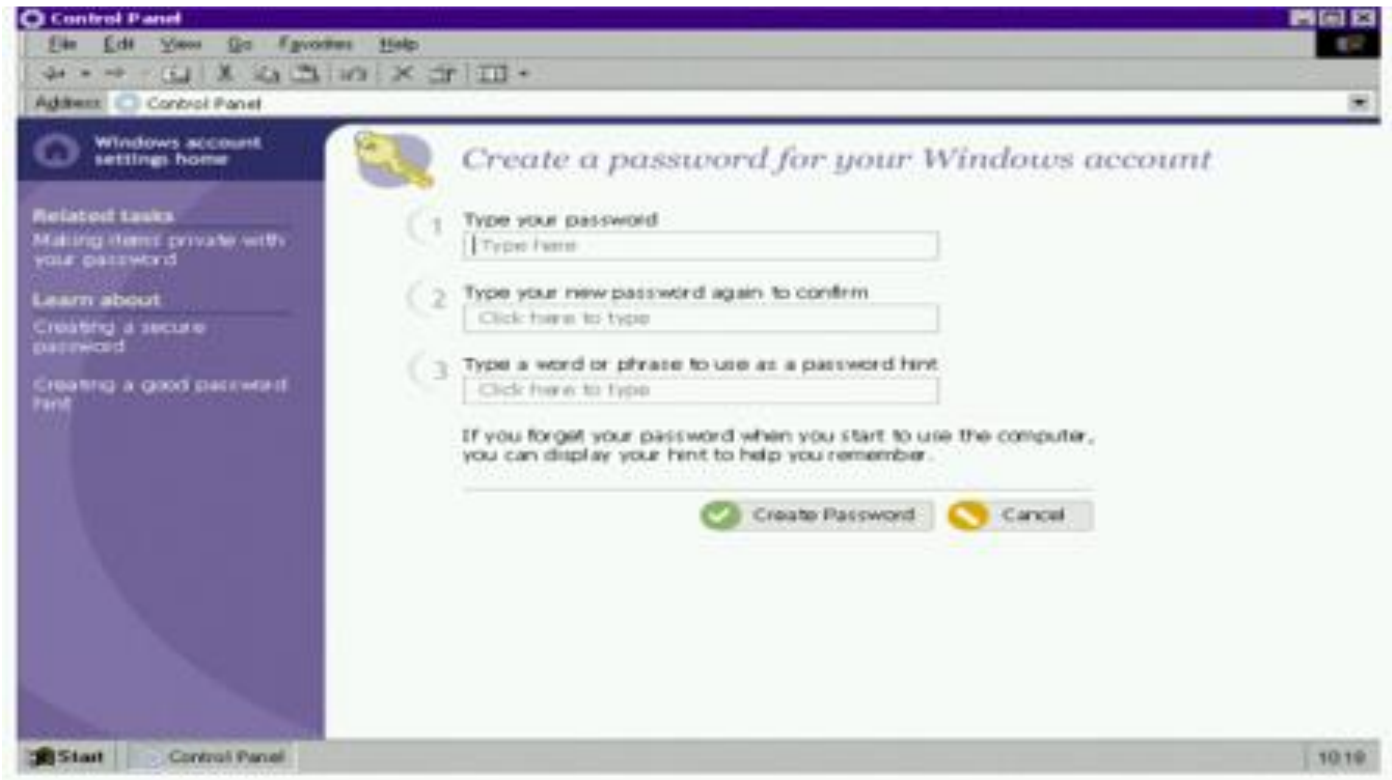
- Money 99 (left) bill payment screen has a vague title - "Upcoming Bills & Deposits“.
- Money 2000 screen on the right, has an explicit title - "Click the bill you'd like to pay“.
- We call a screen-name with no actions a static name, and one with a clear action in it an active one.





Screen title indicates design clarity – cont.

- **An example of screen and title for creating a password in Windows.**



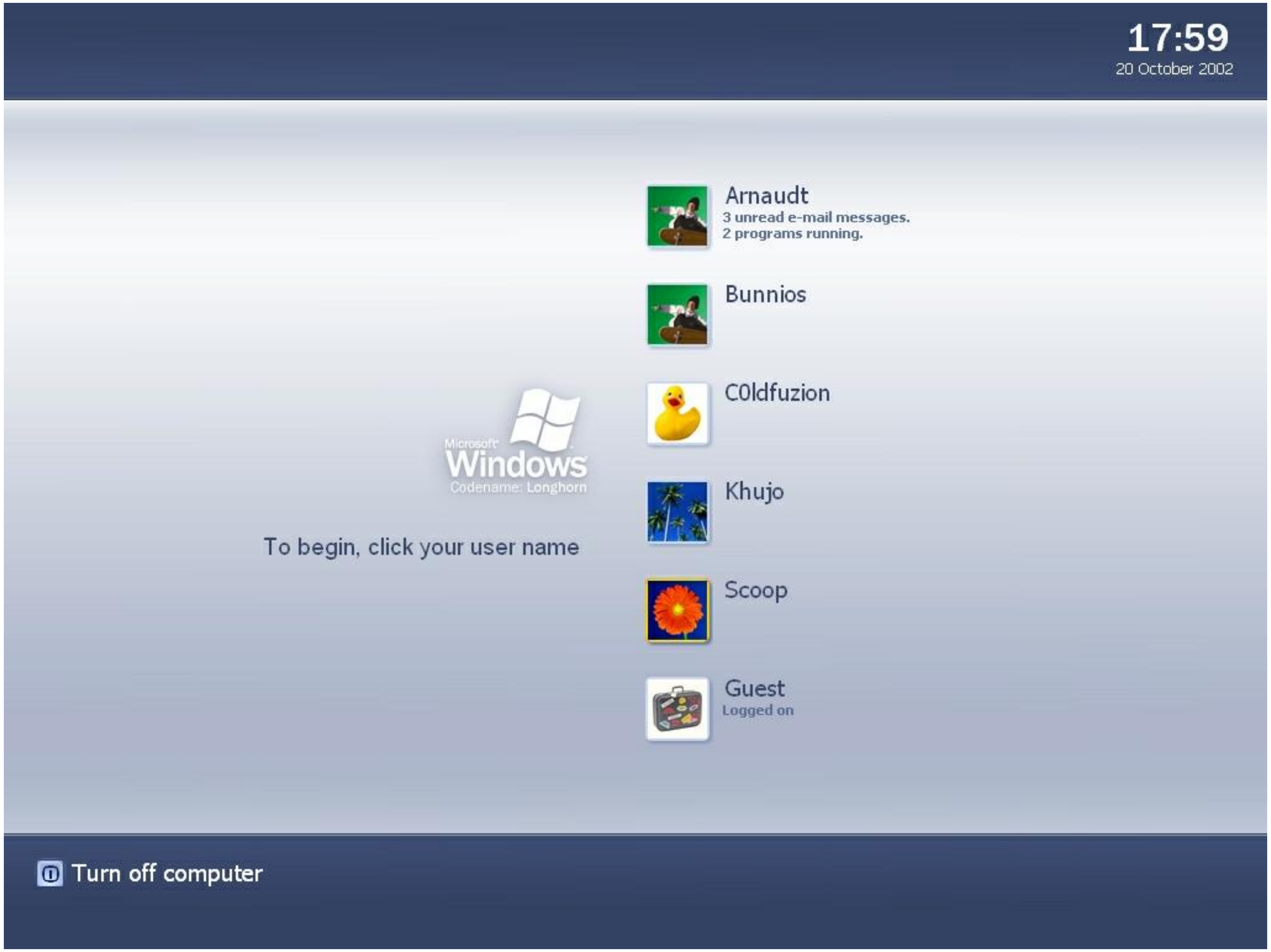
Guidelines for choosing screen titles.

- To use this technique, designers imagine a friend asking, "What is this screen for?"
- If designers can't describe the task without resorting to conjunctions ("and", "or"), the screen is probably trying to do more than one task.
- You can include a brief descriptive paragraph that elaborates on the task.

Step three:
Make the page's contents suit the task.

- After breaking features into screens, the next step is to determine which controls will be used on each screen to accomplish its primary task.
- Examples:
  - "Pick an account to use." – this screen should obviously contain a simple list of accounts the user can choose from.
  - "Check the items to include in your taxes." - naturally, this screen contains a checklist of items.

➡ Justifying the title.



screen contains a simple list of accounts the user can choose from.

Make the page's contents suit the task - cont.

Screen content areas

- In Money 2000, the screen content region is everything below the screen title and to the right of the task list.
- Designers may choose to elaborate on the screen's primary task in a paragraph at the top of the content region.
- If designers want a page to display non-essential reminders, alerts, or other status information, they can be shown to the left of the main content area, under the task list.

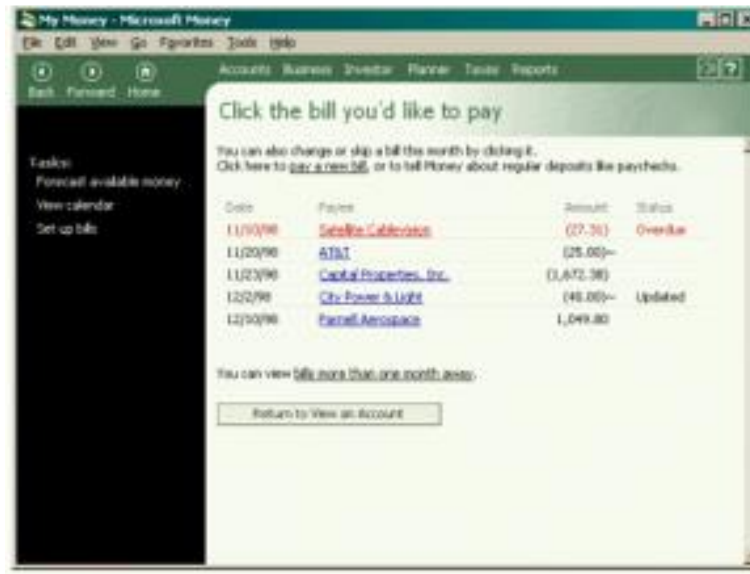
Step four:
Offer links to secondary tasks

- For example, if the primary task on a screen is to write a letter, secondary tasks on that screen might be to look up a mailing address or print an envelope.
- A secondary task might indirectly support the primary task (printing an envelope to send a letter), or might redirect lost users to the place they're searching for (if the current screen doesn't suit their intentions).
- Some screens in your product will have no secondary tasks, while others will have several.

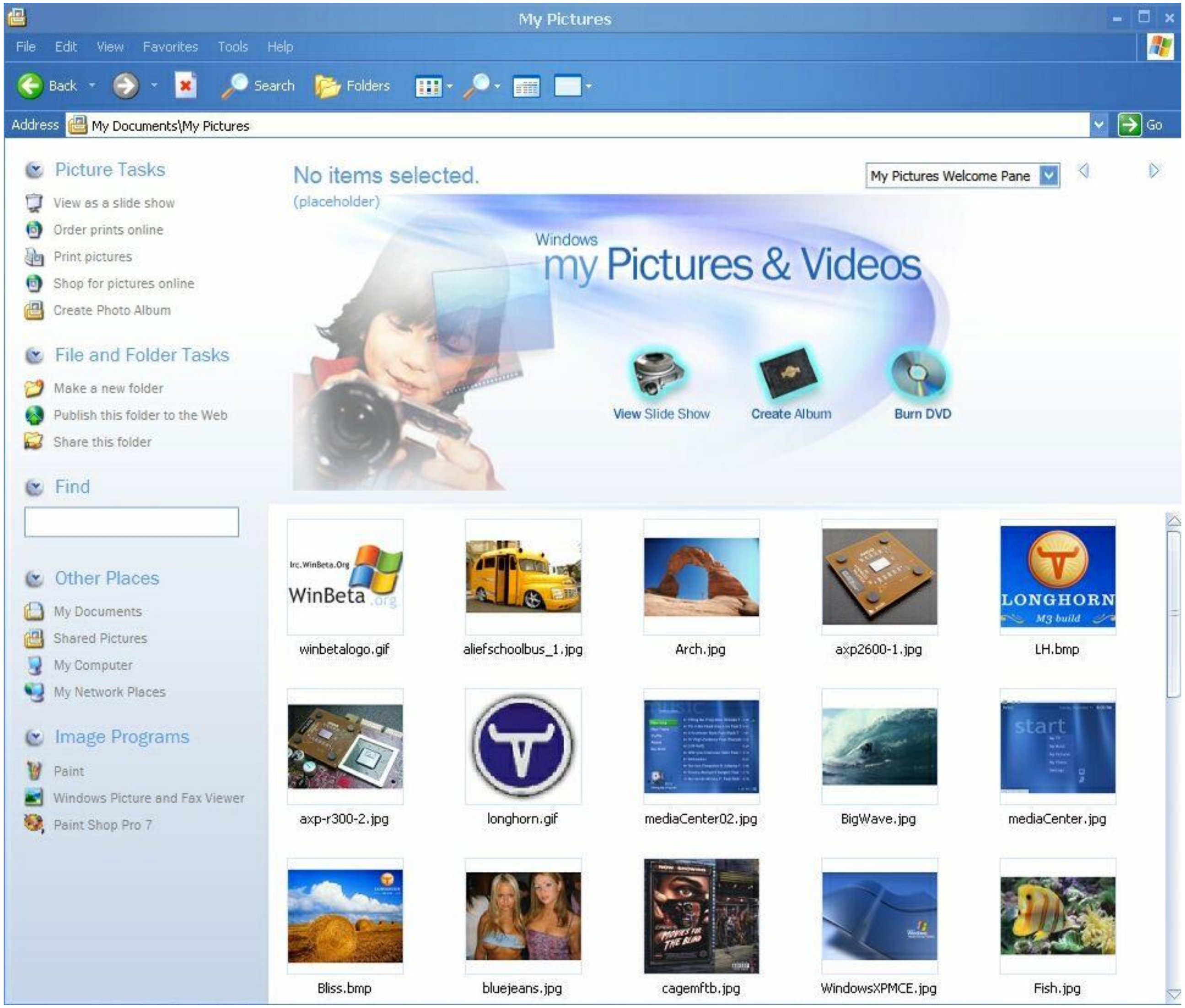
Offer links to secondary tasks – cont.

Visual design

- Secondary tasks should be accessible if needed.
- Must not distract the user from the primary task .
- List should be placed in a consistent location on the screen, in an unscrollable position.



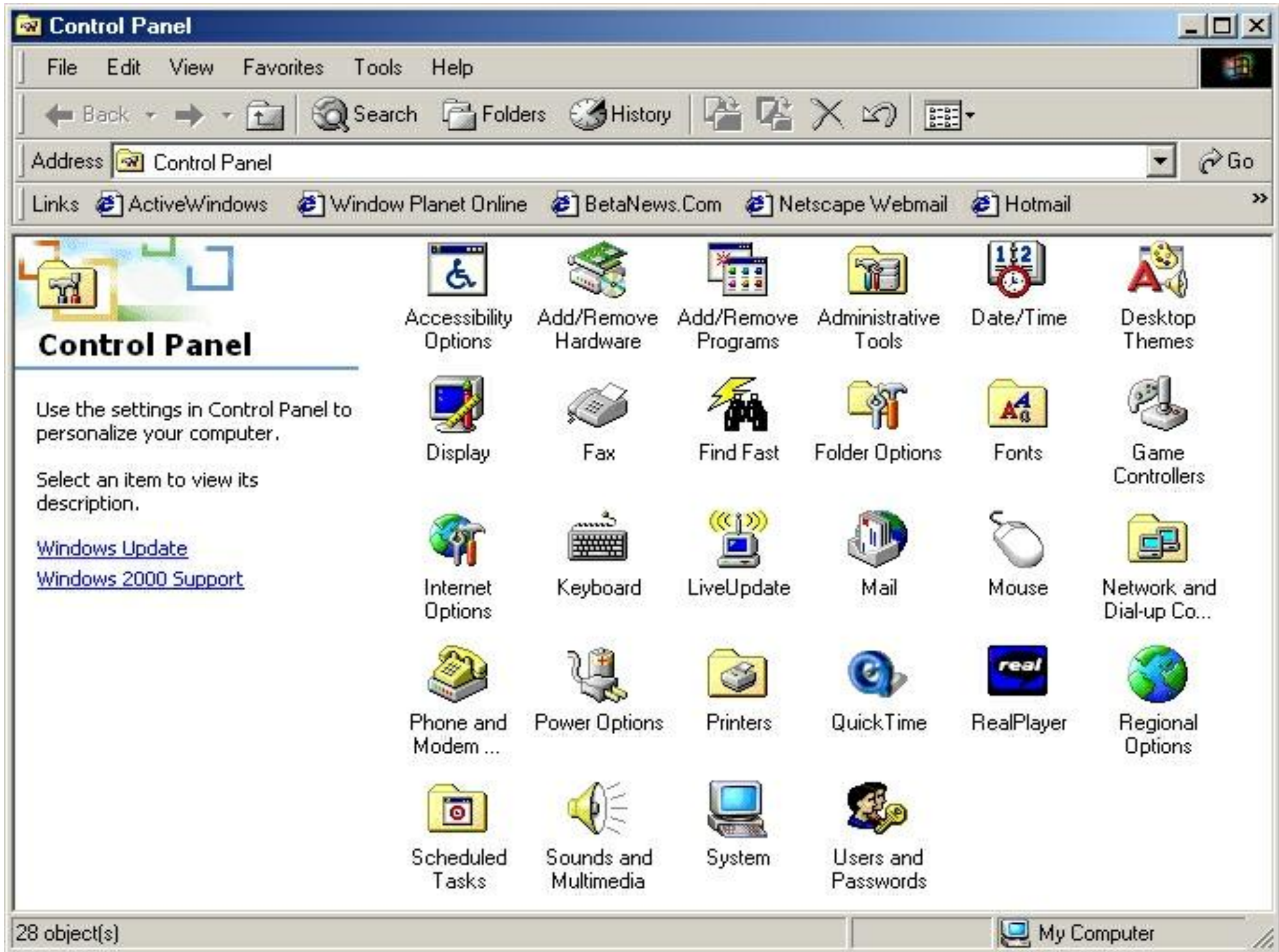
Note the secondary tasks listed on the left



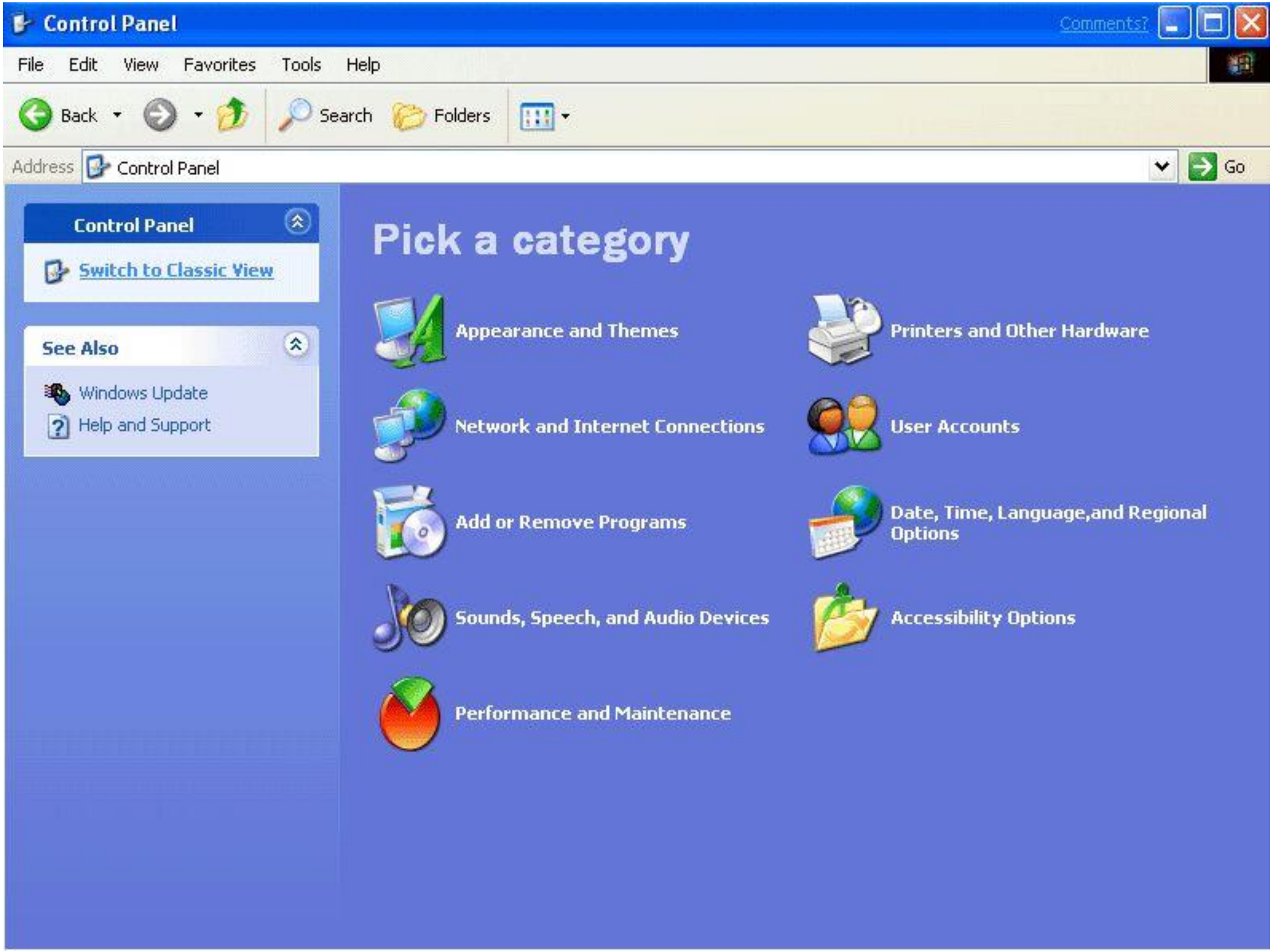
Step four: Offer links to secondary tasks

Provide screens for starting tasks

- Help the user get started by listing groups of common tasks.
- **Activity pages** - organize related groups of common tasks - "What do you want to do now?".
- An activity page makes a good default start page for a product.



Windows 2000 control panel



Windows XP control panel – An activity page

Provide an easy way to complete a task and start a new one

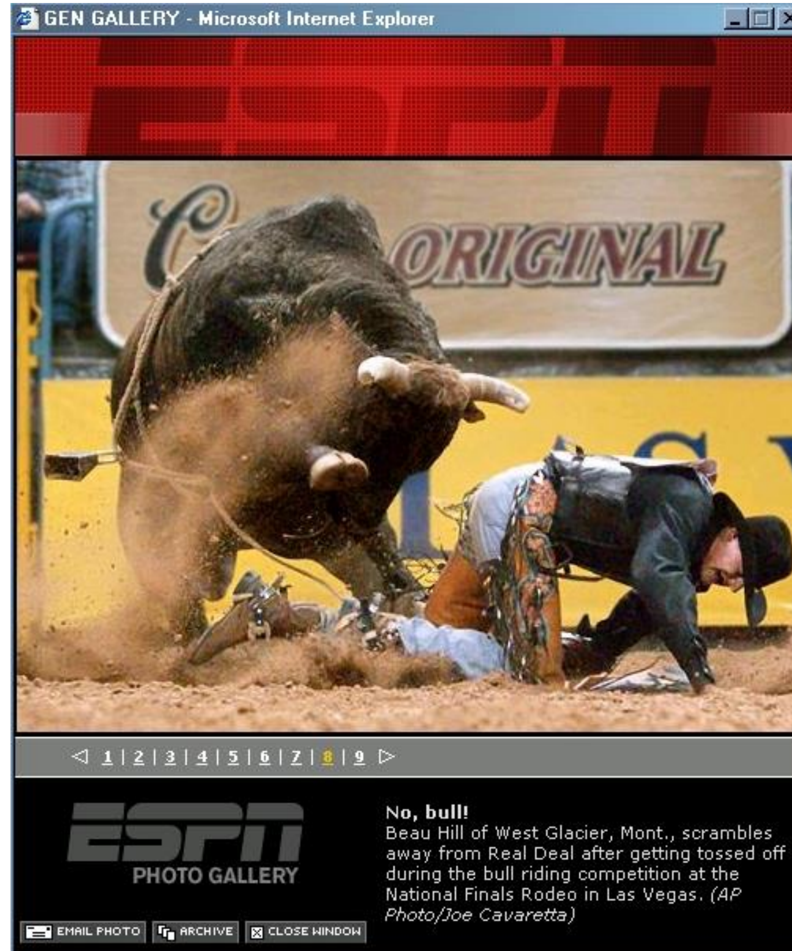
- **Starting and completing processes:** Most users expect to find a button on the final screen that returns them directly to the branching point of the previous process.

- The inductive interface model supports this scenario through the concept of a <u>process</u>, a screen or series of screens treated as a navigational unit.

- **Process name: e**ach process should be given a name, and the name should appear somewhere on each screen in the process.

Provide an easy way to complete a task and start a new one – cont.

- **Done button:** To finish a screen and move on to the next screen in the process, screens can display a button near the bottom of the page.
- **Navigation bar:** A navigation toolbar can offer the user a set of links for starting new tasks.



Notice "close window" button

Make the next navigational Step obvious

- Users are more successful at navigation if they can easily see how to get at least one step closer to their desired result.
- The most frequent tasks should be most prominent and require the least amount of navigation.

How'd they do that?

- Use consistent screen templates
- Provide screens for starting tasks
- Provide an easy way to complete a task and start a new one
- Make the next navigational step obvious

User Assistance

- Guidelines for integrating user assistance text into a product that uses inductive interface.

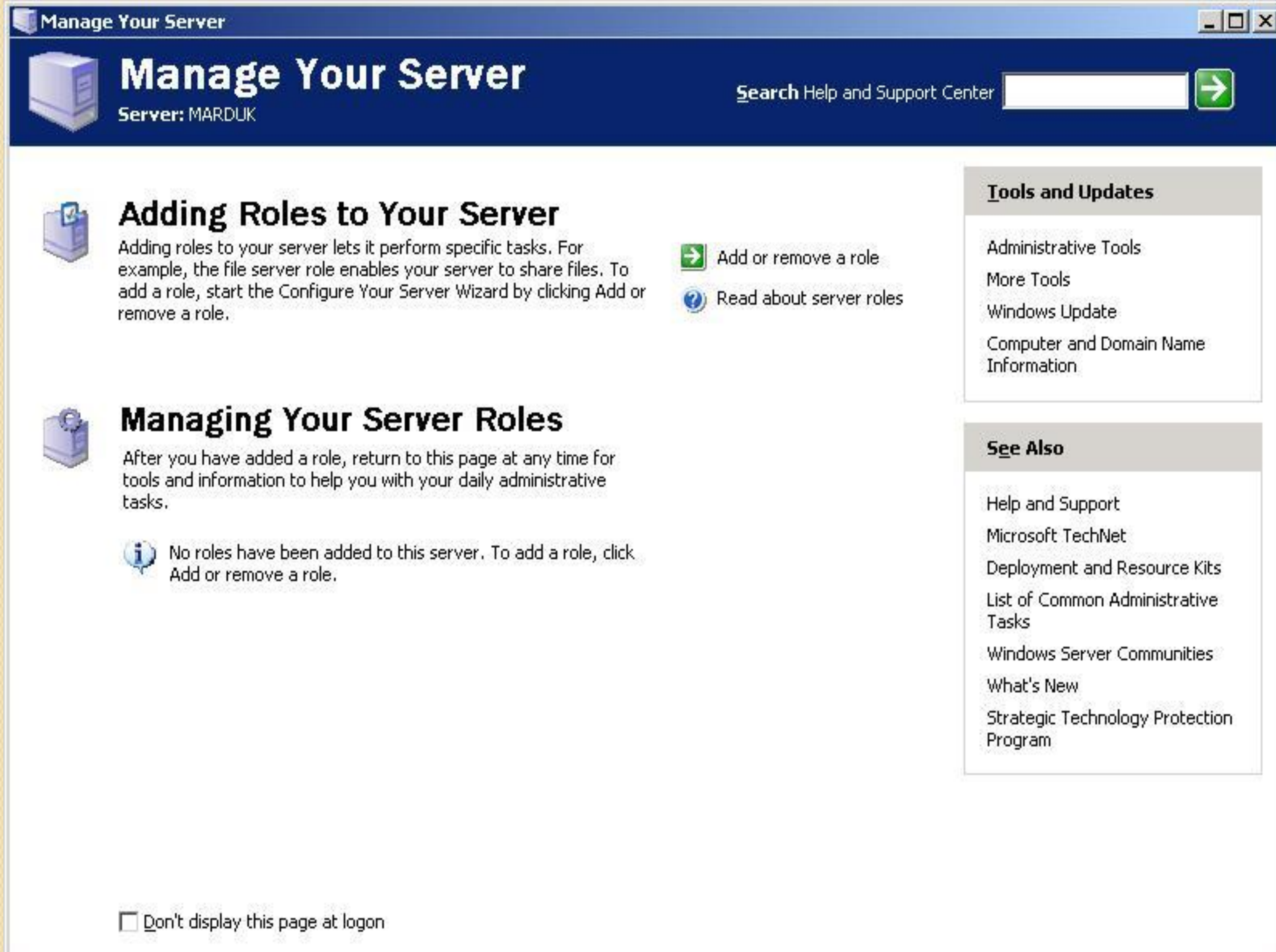
- **Primary assistance:**
  - Text that is visible on the screen.
  - Provides task-focused, textual clues so that users can easily understand all the information presented on the screen.

text is included below each
task link to aid user comprehension.

User Assistance

- **Primary assistance:**
  - **Screen title -** Example: "Change your picture" (user's own language)
  - **Screen subtitle -**
    - Example: You can also download a new picture from the Internet.
    - The subtitle allows you to elaborate on the screen's purpose.
  - **"See also" links – offers links to secondary tasks.**
  - **"Related tasks" links – explicit entry points to other tasks.**

Windows Longhorn display properties – An activity page

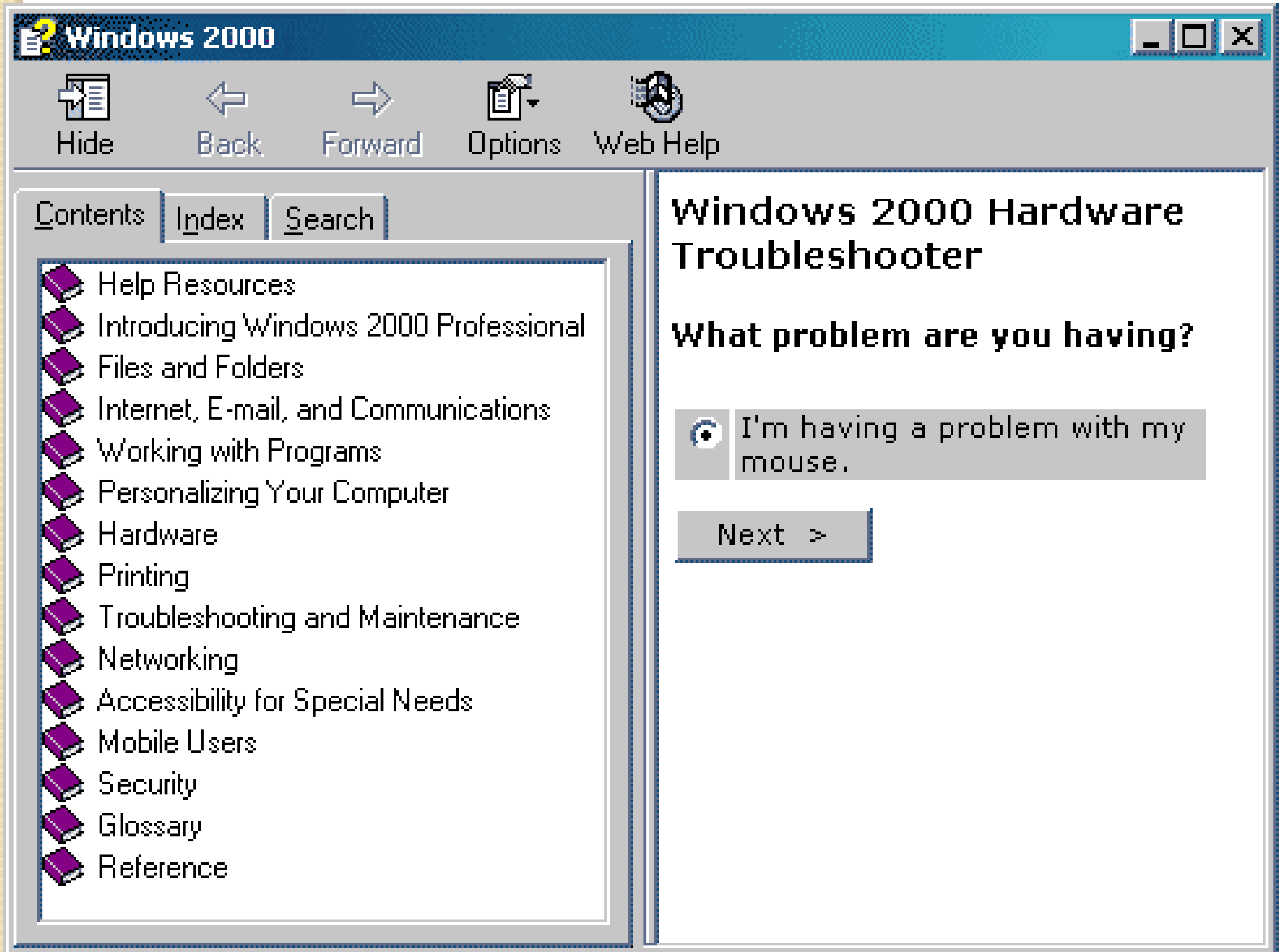


Secondary assistance

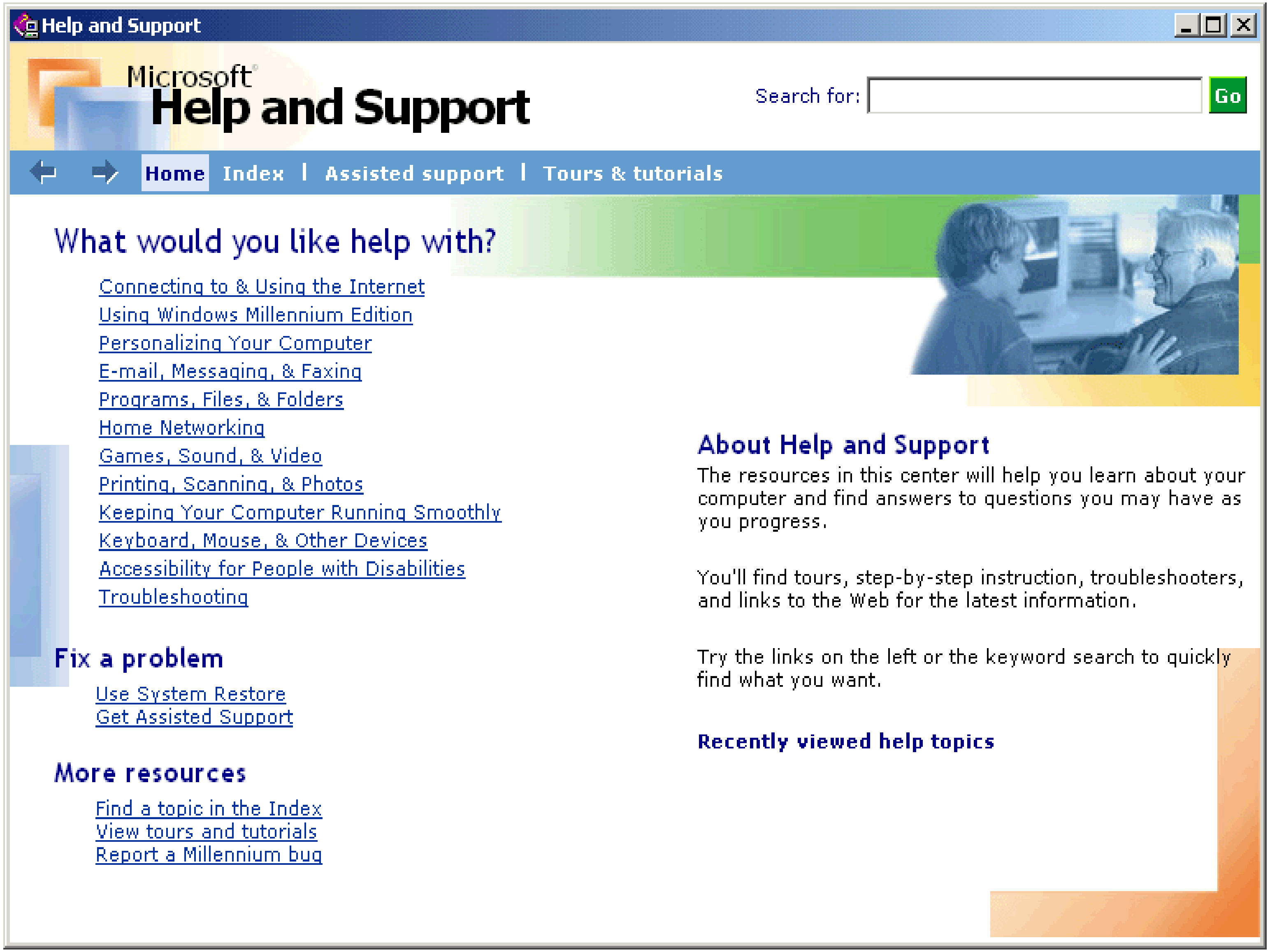
User Assistance

- **Secondary assistance:**
  - consists of all of the text that is not visible on the screen, and requires some user interaction to access, such as clicking or hovering over a user interface element.
  - This content is not essential to accomplish the task at hand, but is still directly related.

- **Examples:**
  - **Info Tips:** appears when the user hovers the mouse over the associated object.
  - **"Learn about" help topics:** open Help topics such as feature overviews, conceptual information, supporting information, and procedural information.



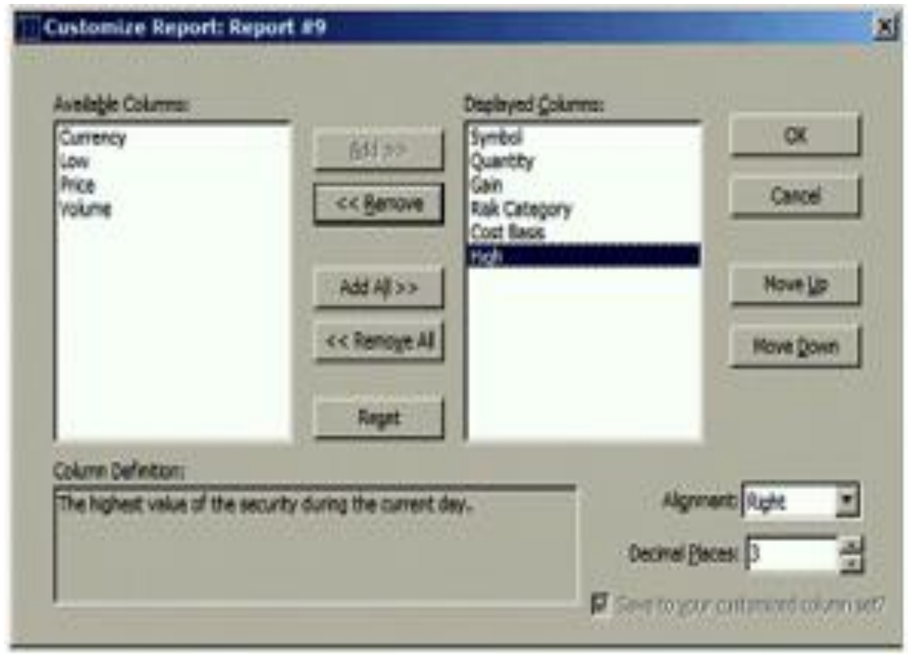
Windows 2000 – Help menu



Windows ME – Help menu

Usability Tests: Controlled testing

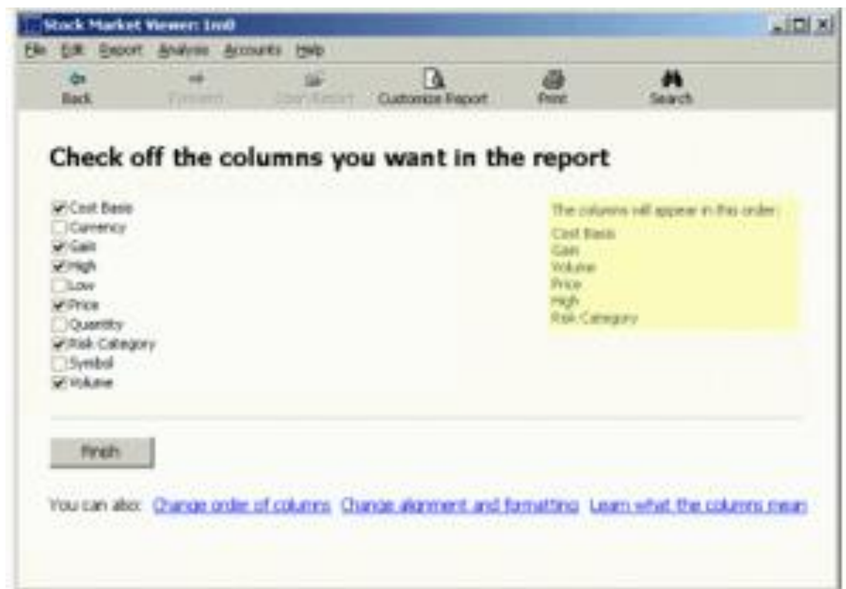
- Microsoft conducted usability tests to examine the difference between money 99 deductive interface and money 2000 inductive interface model.
- subjects were asked to customize reports from a given starting state to a specified goal state in one of the tested interfaces.
- See next foil for the inductive interface version of this form.



conventional user interface dialog was used for selecting and manipulating report columns.

Controlled testing – cont.

- **inductive interface version of test application:**



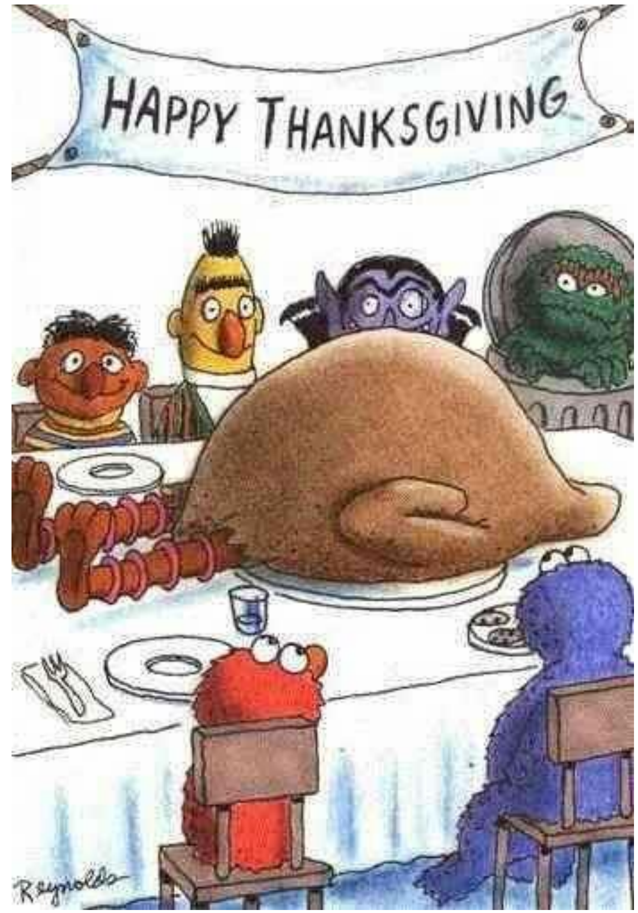
- **Each page performs a single task that is clearly stated in the page's title.**

Conclusions - Usability Tests

- There were no significant differences between users' performances in the two models or any of the other variables of interest.
- Users performed tasks at the same speed, iterated on the task the same number of times, and had the same overall subjective satisfaction ratings for the two versions.
- **This test, therefore, failed to show any measures in which inductive interface resulted in an improvement or degradation in performance or subjective ratings.**

Conclusions - Usability Tests – cont.

- Further research will be necessary to determine if there are measurable improvements from the use of inductive interface.
- At the very least, this test did not provide any evidence that inductive interface harms performance or product usage.



The Down Side of inductive interface

The down side of inductive interface

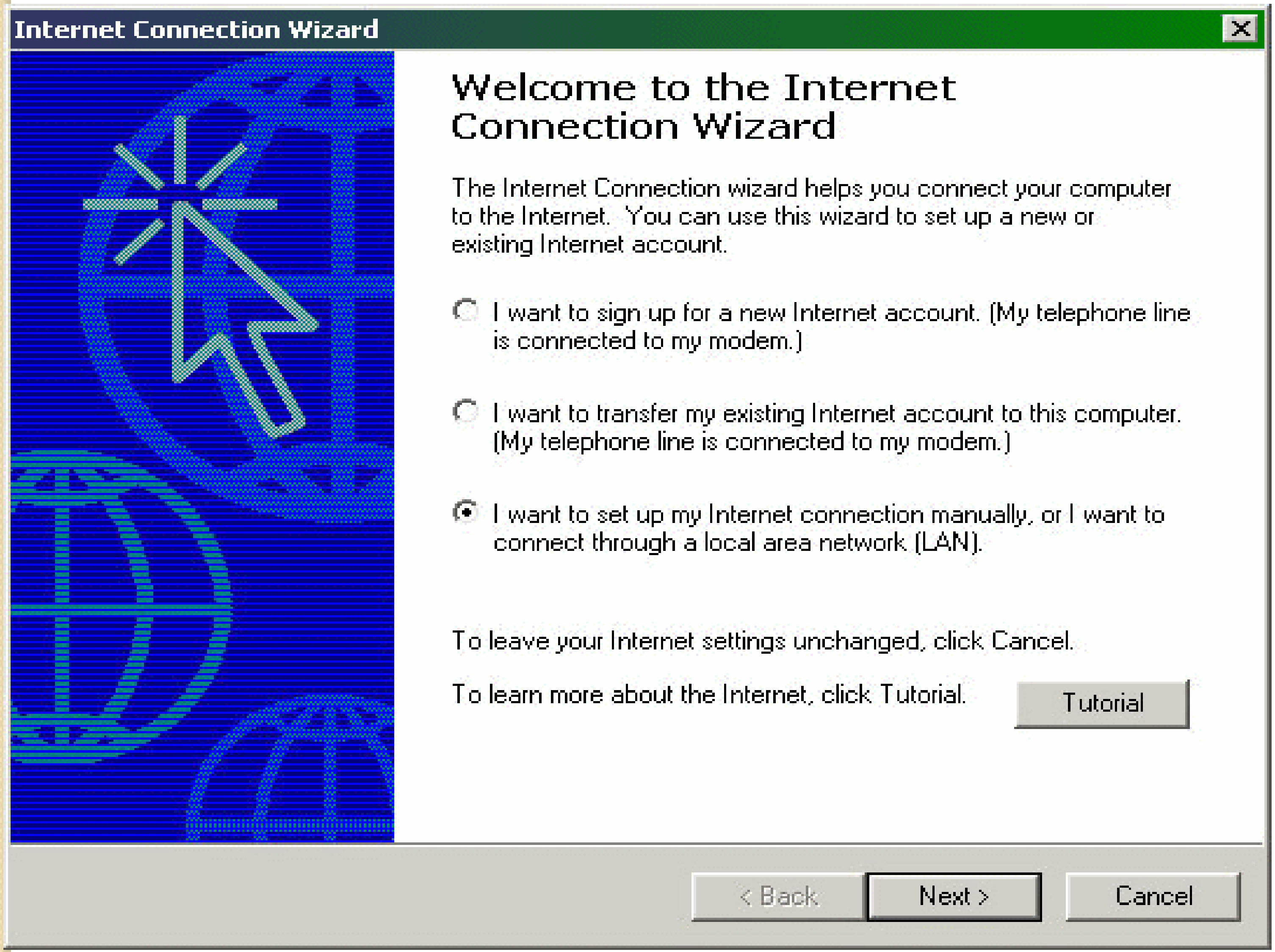
- Bigger programs/software.
- Assumes beginner level at all times.
- Focuses on an individual task
- Hard to implement multi-dimensional tasks.
- Choosing a simple name for a screen isn’t useful in some cases.

Bigger Programs/Software

- Programs implemented using **inductive interface require more screens** per task than any other program.
- Larger amount of screens demands **larger disk space** for any given program.
- Software might become slower due to **bigger memory requirements**.

Assumes beginner level at all times

- "Wizard" type interface which is annoying.
- There's no encapsulation of several concepts on one page.
- Extra clicking is disturbing for experienced users.
- Assumes a user doesn't keep knowledge from previous screen on to this screen.



Yet another annyoing wizard interface

An Example…Hotmail

- A popular program.
- Used by various users, including:
  - Users with no previous knowledge in computers.
  - Users with no computer at all.

Hotmail Inbox

Hotmail | Home | Inbox | Compose | Contacts | Options | Help

GeorgeW@hotmail.com

Free Newsletters | MSN Featured Offers | Find Message

Show me mail from: My Favorites !

<< Hide Folders

- Inbox (21)
- Junk Mail (3)
- Sent Messages
- Drafts
- Trash Can
- Sharon (2789)
- God (2,145,980)
- Rumsfeld (150)
- Dick (78)
- Condy (89)
- Daddy (201)
- Bin Laden (267)
- Arnold (15)

Delete | Block | Mark As Unread | Put in Folder...

From	Subject	Date	Size
Rumsfeld	Plans to invade Iraq are ready, but that spoil-sport Powell disagrees	Sep 1	10k
Hamid Karazai	We need more prostitutes in Kabul, too many GI's around	Sep 1	9k
Ben Ladin	I dare you to open the attachment	Sep 1	260M
Arafat	I am waiting for an invitation to your ranch	Sep 1	1k
Clinton	If you invite Arafat don't let him kiss you	Sep 2	13k
Arafat	I gave instructions in Arabic to stop suiciders	Sep 2	19k
Arafat	I am still waiting for your invitation	Sep 2	1k
Arafat	If you invite me I promise not to kiss you	Sep 2	10k
Condolezza Rice	Arafat is spamming your email, don't answer him	Sep 2	9k
Bin Ladin	You coward you still don't dare to open my email !	Sep 2	261M
Mohamed Ata	You think I'm stupid to kill myself, try and catch me	Sep 3	47k
Arafat	I also promise not to kiss Mrs. Bush	Sep 3	1k
Aschcroft	We caught a terrorist with a nuclear bomb in his crotch?	Sep 3	10k
Rumsfeld	When are we going after Iraq? Wolfowitz is pressuring me	Sep 3	1k
Powell	I have to see you, am I still Secretary of State?	Sep 4	1k
Sharon	It looks like I am entering Gaza before you attack Iraq	Sep 4	1k
Kofi Anan	I don't seem to have any work to do, shall we close down the UN?	Sep 4	6k

The main screen encapsulates many features:

- The inbox lists emails.
- Clicking on a subject takes you to that email.
- Clicking a checkbox enables you to delete checked mail, or block the addresses from which they came from.
- You can jump to a different folder.
- Perform various other tasks.

Safe to say that Hotmail is fairly usable.

inductive interface Implemented Hotmail…

Select a Task to Perform:

- Delete e-Mails.
- Block e-Mails.
- Mark as un-read.
- Move e-Mails.

Next

inductive interface implemented Hotmail…cont.

Select the e-Mails You Wish to Move:

- e-Mail # 1.
- e-Mail # 2.
- e-Mail # 3.
- e-Mail # 4.

Next

inductive interface implemented Hotmail…cont

<u>Select the Folder You Wish to Move these e-Mails to:</u>

Folder # 1.

Folder # 2.

Folder # 3.

Folder # 4.

Next

inductive interface implemented Hotmail…cont

You Have Placed X e-Mails in Folder # 2.

Done

Focuses on an individual task

- Disagrees with the “real world” model of objects.
- Hard to get overview of the system.
- No sense of hierarchy, parallel functions, etc.
- Users are likely to only know a small percent of all the functions in the program.

An Example…A Stereo System

- Easy to get overview of the system.
- All parallel functions are visible at any given time.
- Easier to operate.

- Could you imagine a screen for each one of the stereo system component?

Focuses on an individual task – cont.

- Easy to get lost in inductive interface .
- Every time you perform a single task you are required to re-orient yourself.

An Example…Driving Directions

Here are some driving directions:

1. Go down this road.
2. Turn left after the gas station.
3. Pass a bunch of trees.
4. Turn right at the second traffic light.
5. Continue 2 miles and turn left.
6. Pass the public school…

And you're there…

5 minutes later…

What went wrong?

- The person giving you the directions has an internal map and can orient to it.
- One wrong turn will put you in an unfamiliar place without any way of re-orienting yourself.
- No way of knowing what went wrong.
- You have to make a U turn and approach the whole thing again.

Isn't This Better…?

Hard to implement multi -dimensional tasks

- Tasks that effect each another.
- Tasks that contain many variables and their interaction.

A Simple Example…

Reflections on inductive interface

- Pro's:
  - Simple
  - practical.
  - Extremely good for beginners
  - Welcomed innovation
- Con's:
  - Annoying for the experienced users
  - Hard to get overview of the system

When to use

- Should be used in moderation – 100% inductive interface software is just too much.
- Useful in infrequently used tasks, where the user is most likely to forget how to accomplish them.
- Useful in infrequently used systems, such as ATM's and vending machines…

References

- **"Microsoft Inductive User Interface Guidelines".**

  Microsoft Corporation, February 9, 2001
- **"Why technical writers should love Microsoft's Inductive User Interface".**

  Janice Carlson
- **"why I don't like the Inductive user interface"**

  posts on "Joel on software Forum" about why I don't like the inductive interface
- **Screenshots and pictures**

  The Internet

- http://start-jandroid.blogspot.com/2011/01/android-multiple-screen-example.html

- http://developer.android.com/resources/tutorials/notepad/notepad-ex1.html